

Task 4 – System Hardening

Objective:

Perform basic system hardening by applying updates, configuring firewall rules using iptables, and disabling an unused service.

1 – System Update:

Executed `sudo apt update` to refresh package repositories.

```
kali linux 2025 (Running) - Oracle VM VirtualBox
File Machine View Input Devices Help
mudhiraj@kali: ~/Desktop/windows_files/malware_basics

(mudhiraj@kali)~-[~/Desktop/windows_files/malware_basics]
$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.3 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.4 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [104 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [189 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [175 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [891 kB]
Fetched 1,359 kB in 6s (235 kB/s)
136 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

2 – Package Upgrade:

Executed `sudo apt upgrade -y` to install available security updates.

```
kali linux 2025 (Running) - Oracle VM VirtualBox
File Machine View Input Devices Help
mudhiraj@kali: ~/Desktop/windows_files/malware_basics

(mudhiraj@kali)~-[~/Desktop/windows_files/malware_basics]
$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  libabsl20240722 libmjpegutils-2.1-0t64 libmpeg2encpp-2.1-0t64 libmplex2-2.1-0t64 libvidstab1.1 libwireshark18 libwiretap15 libwsutil16 libximg3 n
Use 'sudo apt autoremove' to remove them.

Upgrading:
adwaita-icon-theme gcc-15-base kali-tools-top10 libgprofng0 libprotobuf-lite32t64 node-interpre
alsa-ucm-conf gcc-15-for-host libavcodec62 libgstreamer-plugins-bad1.0-0 libprotobuf32t64 node-postcss
atftpd gcc-15-x86-64-linux-gnu libavfilter11 libhavege2 libprotoc32t64 offsec-awae-python2
binutils gettext-base libavformat62 libidn12 libsframe3 pipewire
binutils-common girl.2-packagekitglib-1.0 libavutil60 libjsoncpp26 libspa-0.2-bluetooth pipewire-bin
binutils-x86-64-linux-gnu glycin-loaders libbinutils liblilv-0-0 libspa-0.2-modules pipewire-pulse
clang-19 glycin-thumbnaillers libbtbb1 libllvm19 libstdc++-15-dev python3-aiohttp
clang-tools-19 gstreamer1.0-libav libclang-common-19-dev liblua5.4-0 libswresample6 python3-anyio
cpp-15 gstreamer1.0-plugins-bad libclang-cpp19 liblua5.5-0 libswscale9 python3-cbor2
cpp-15-for-host gstreamer1.0-plugins-good libclang-rt-19-dev libmongocrypt0 libtirff6 python3-elastic-transpo
cpp-15-x86-64-linux-gnu haveged libclang1-19 libobjc-15-dev libwinpr3-3 python3-filelock
docbook+xml kali-desktop-core libctf-nobfd0 libpackagekit-glib2-18 python3-itsdangerous
firefox-esr kali-desktop-xfce libctf0 libpam-modules liblvm-19 python3-mpmath
freerdp-x11 kali-linux-core libfluidsynth3 libpam-modules-bin libllvm-19-dev python3-pyramid
freerdp3-x11 kali-linux-firmware libfreerdp-client3-3 libpam-runtime liblvm-19-runtime python3-redis
g++-15 kali-system-cli libfreerdp3-3 libpam0g liblvm-19-tools python3-tqdm
g++-15-x86-64-linux-gnu libgcc-15-dev libpipewire-0.3-0t64 mesa-vulkan-drivers python3-tzlocal
gcc-15 kali-system-gui libglycin-2-0 libpipewire-0.3-modules node-builtins python3-virtualenv

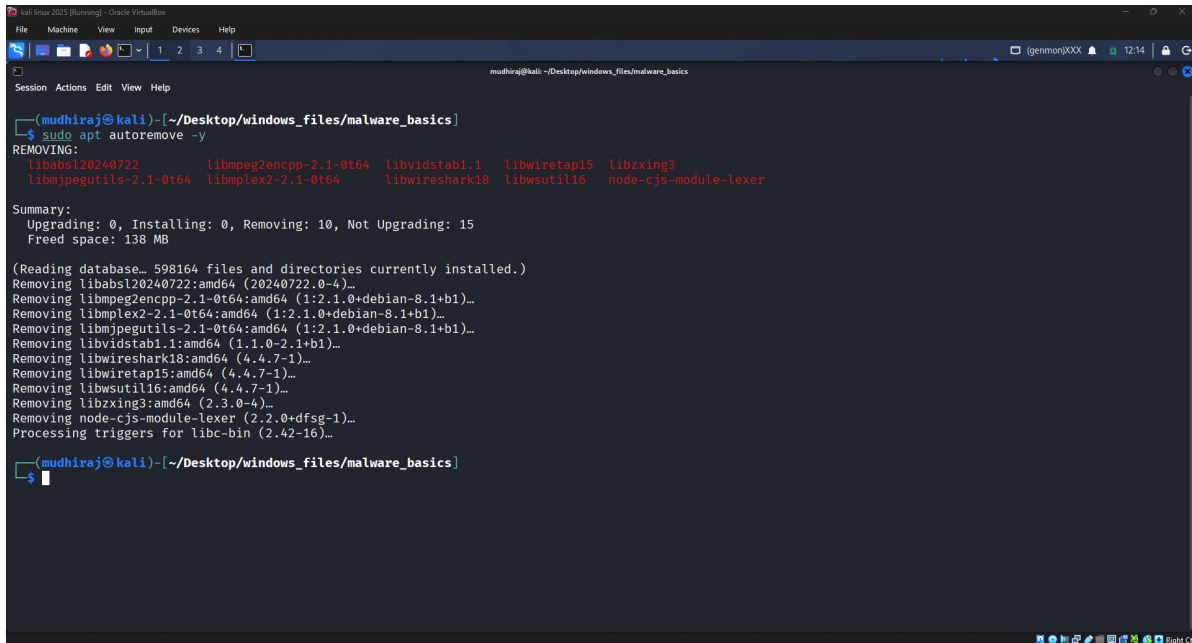
Installing dependencies:
libvidstab1.2

Not upgrading:
colord libcolorhug2 libgbm1 libglx-mesa0 libgtk-4-bin libpskc0t64 node-shebang-command yelp
graphviz libegl-mesa0 libgl1-mesa-dri libgtk-4-1 libgtkmm-4.0-0 mesa-libgallium python3-pygraphviz

Summary:
Upgrading: 121, Installing: 1, Removing: 0, Not Upgrading: 15
```

3 – Remove Unused Packages:

Executed `sudo apt autoremove -y` to remove unnecessary packages.



```
(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$ sudo apt autoremove -y
REMOVING:
libabsl20240722 libmpeg2encpp-2.1-0t64 libvidstab1.1 libwiretap15 libzxing3
libmjpegutils-2.1-0t64 libplex2-2.1-0t64 libwireshark18 libwsutil16 node-cjs-module-lexer

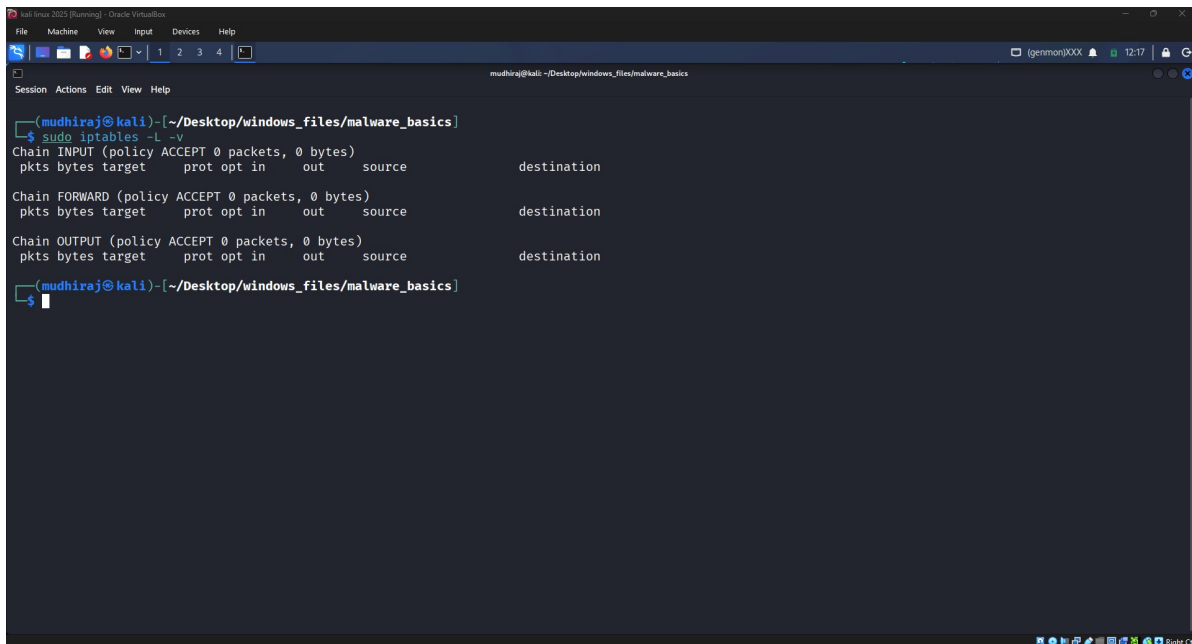
Summary:
Upgrading: 0, Installing: 0, Removing: 10, Not Upgrading: 15
Freed space: 138 MB

(Reading database, 598164 files and directories currently installed.)
Removing libabsl20240722:amd64 (20240722.0-4)...
Removing libmpeg2encpp-2.1-0t64:amd64 (1:2.1.0+debian-8.1+b1)...
Removing libplex2-2.1-0t64:amd64 (1:2.1.0+debian-8.1+b1)...
Removing libmjpegutils-2.1-0t64:amd64 (1:2.1.0+debian-8.1+b1)...
Removing libvidstab1.1:amd64 (1.1.0-2.1+b1)...
Removing libwireshark18:amd64 (4.4.7-1)...
Removing libwiretap15:amd64 (4.4.7-1)...
Removing libwsutil16:amd64 (4.4.7-1)...
Removing libzxing3:amd64 (2.3.0-4)...
Removing node-cjs-module-lexer (2.2.0+dfsg-1)...
Processing triggers for libc-bin (2.42-16)...

(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$
```

4 – Initial Firewall Rules:

Displayed the current iptables rules using `sudo iptables -L -v`.



```
(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source    destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source    destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source    destination

(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$
```

5 – Allow SSH:

Added a rule allowing inbound SSH traffic (TCP port 22).

```
kali linux 2023 (Running) - Oracle VM VirtualBox
File Machine View Input Devices Help
mudhiraj@kali: ~/Desktop/windows_files/malware_basics

(mudhiraj@kali)~/Desktop/windows_files/malware_basics
$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination

(mudhiraj@kali)~/Desktop/windows_files/malware_basics
$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

(mudhiraj@kali)~/Desktop/windows_files/malware_basics
$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination
0      0 ACCEPT      tcp  --  any    any    anywhere          anywhere          tcp dpt:ssh

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination

(mudhiraj@kali)~/Desktop/windows_files/malware_basics
$
```

6 – Block Telnet:

Added a DROP rule for Telnet (TCP port 23) and verified with line numbers.

```
kali linux 2023 (Running) - Oracle VM VirtualBox
File Machine View Input Devices Help
mudhiraj@kali: ~/Desktop/windows_files/malware_basics

(mudhiraj@kali)~/Desktop/windows_files/malware_basics
$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

(mudhiraj@kali)~/Desktop/windows_files/malware_basics
$ sudo iptables -L -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination
0      0 ACCEPT      tcp  --  any    any    anywhere          anywhere          tcp dpt:ssh

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination

(mudhiraj@kali)~/Desktop/windows_files/malware_basics
$ sudo iptables -A INPUT -p tcp --dport 23 -j DROP

(mudhiraj@kali)~/Desktop/windows_files/malware_basics
$ sudo iptables -L -v --line-numbers
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target      prot opt in     out     source            destination
1    0      0 ACCEPT      tcp  --  any    any    anywhere          anywhere          tcp dpt:ssh
2    0      0 DROP        tcp  --  any    any    anywhere          anywhere          tcp dpt:telnet

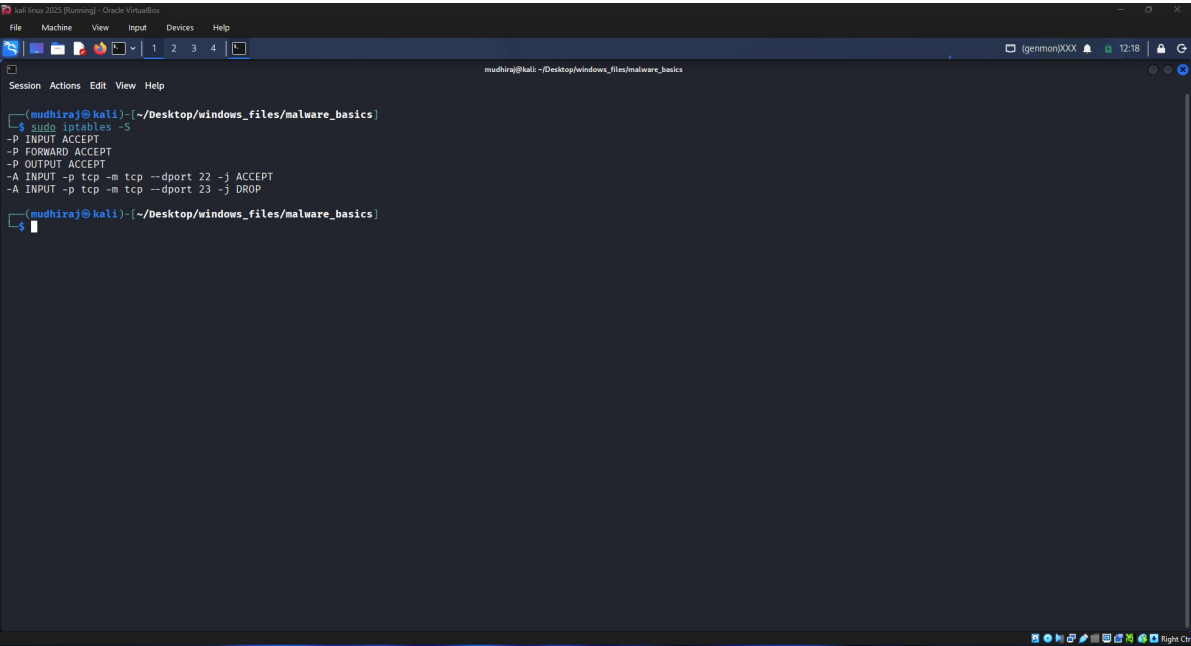
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target      prot opt in     out     source            destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target      prot opt in     out     source            destination

(mudhiraj@kali)~/Desktop/windows_files/malware_basics
$
```

7 – Firewall Rule Summary:

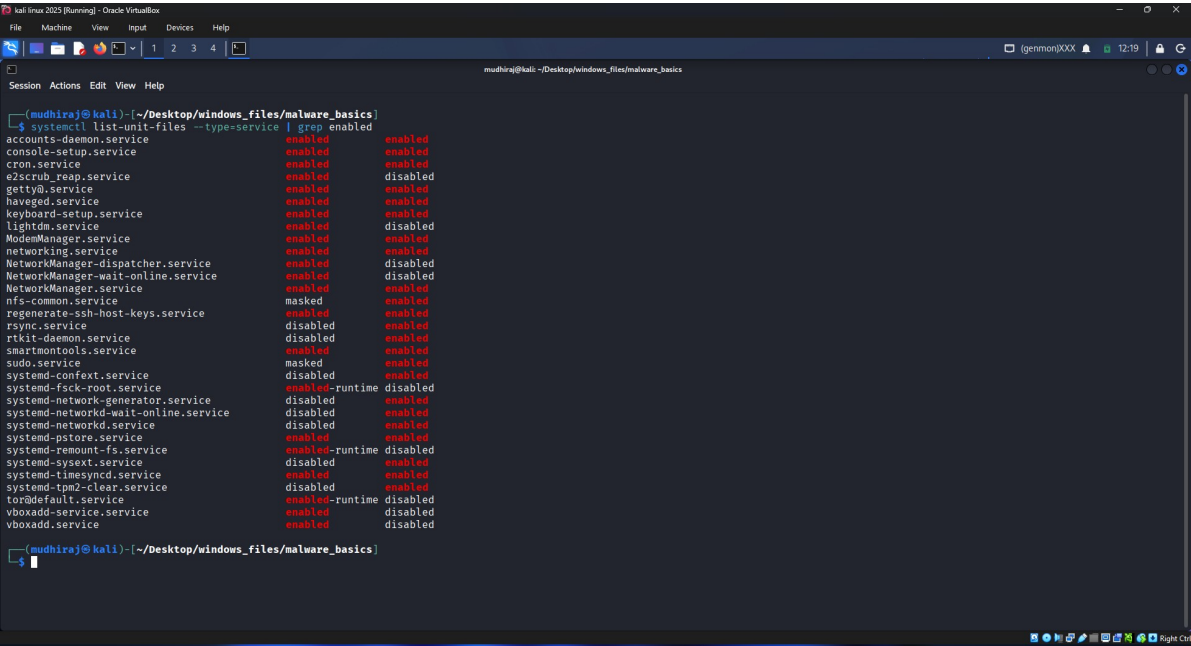
Verified firewall rules using `sudo iptables -S`.



```
(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$ sudo iptables -S
-P INPUT ACCEPT
-P FORWARD ACCEPT
-P OUTPUT ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 23 -j DROP
(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$
```

8 – Enabled Services:

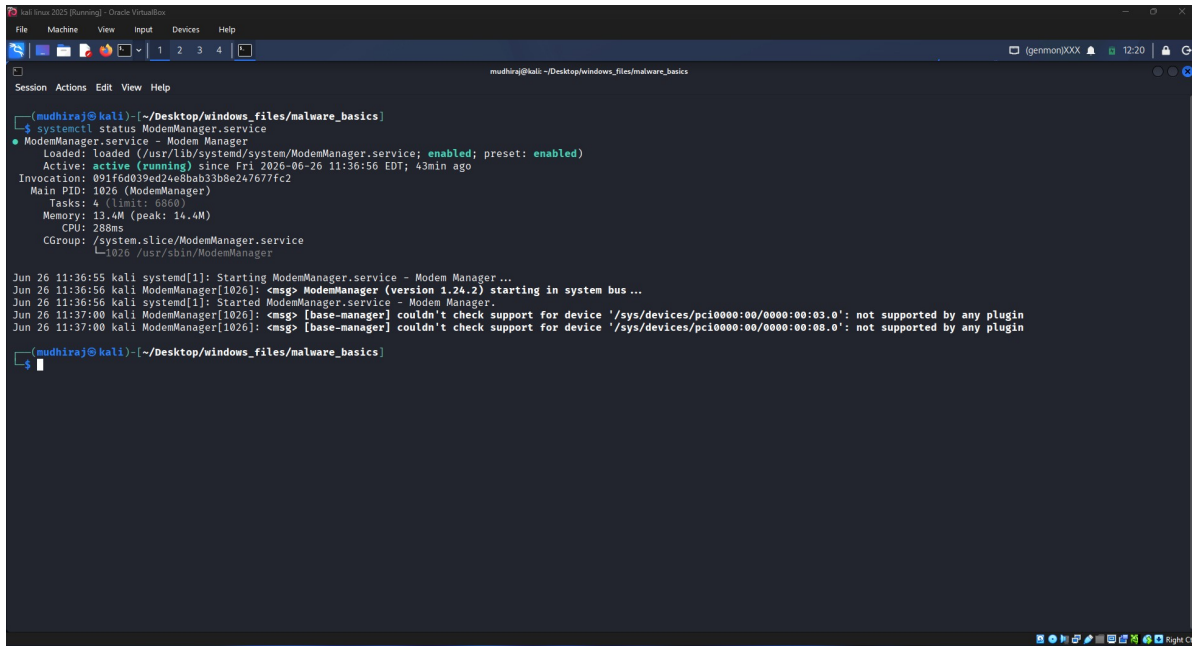
Listed enabled system services to identify unused services.



```
(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$ systemctl list-unit-files --type=service | grep enabled
accounts-daemon.service          enabled enabled
console-setup.service           enabled enabled
cron.service                     enabled enabled
e2scrub_reap.service            enabled disabled
getty@.service                  enabled enabled
haveged.service                 enabled enabled
keyboard-setup.service          enabled enabled
lightdm.service                 enabled disabled
ModemManager.service           enabled enabled
networking.service             enabled enabled
NetworkManager-dispatcher.service disabled disabled
NetworkManager-wait-online.service disabled disabled
NetworkManager.service         enabled enabled
nfs-common.service             masked enabled
regenerate-ssh-host-keys.service disabled enabled
rsync.service                   disabled enabled
rtkit-daemon.service           disabled enabled
smartmontools.service          enabled enabled
sudo.service                   masked enabled
systemd-conftest.service        disabled enabled
systemd-fsck-root.service       enabled-runtime disabled
systemd-network-generator.service disabled enabled
systemd-networkd-wait-online.service disabled enabled
systemd-networkd.service        disabled enabled
systemd-pstore.service          enabled enabled
systemd-remount-fs.service      enabled-runtime disabled
systemd-sysext.service          disabled enabled
systemd-timesyncd.service       enabled enabled
systemd-tmp2-clear.service      disabled enabled
tor@default.service            enabled-runtime disabled
vboxadd-service.service         enabled disabled
vboxadd.service                enabled disabled
(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$
```

9 – ModemManager Status:

Checked the status of ModemManager before disabling it.



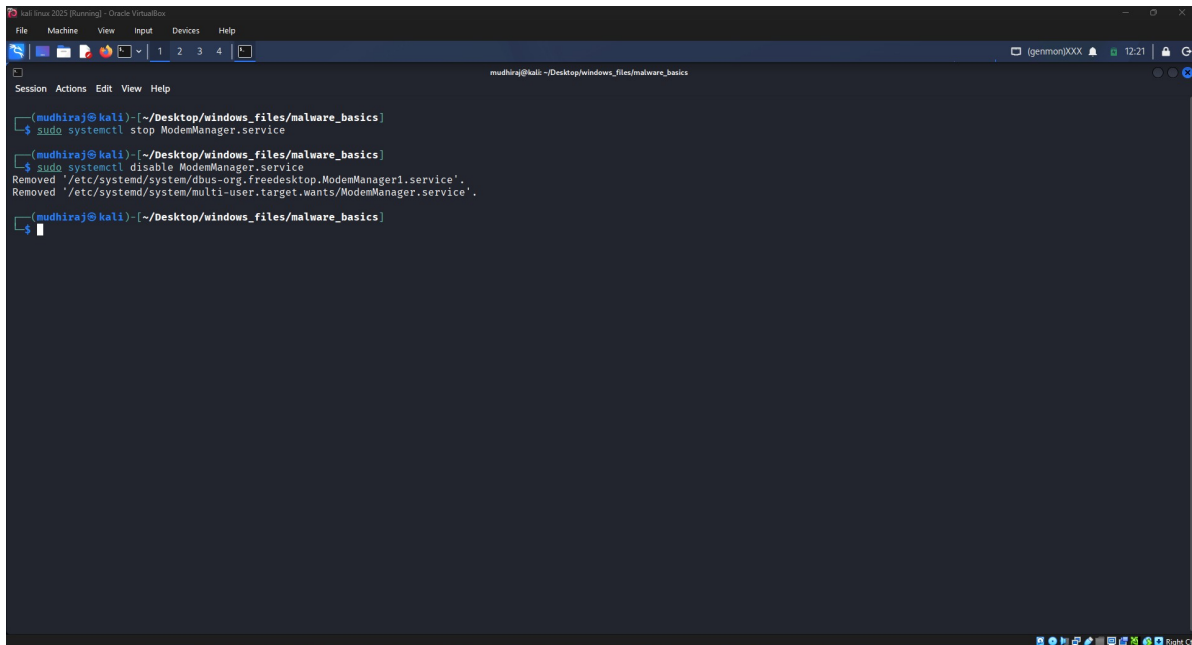
```
(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$ systemctl status ModemManager.service
● ModemManager.service - Modem Manager
   Loaded: loaded (/usr/lib/systemd/system/ModemManager.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-06-26 11:36:56 EDT; 43min ago
  Invocation: 091f6d039ed24e8bab33b8e247677fc2
    Main PID: 1026 (ModemManager)
      Tasks: 4 (limit: 6553)
     Memory: 13.4M (peak: 14.4M)
        CPU: 288ms
    CGroup: /system.slice/ModemManager.service
            └─1026 /usr/sbin/ModemManager

Jun 26 11:36:55 kali systemd[1]: Starting ModemManager.service - Modem Manager...
Jun 26 11:36:56 kali ModemManager[1026]: <msg> ModemManager (version 1.24.2) starting in system bus ...
Jun 26 11:36:56 kali systemd[1]: Started ModemManager.service - Modem Manager.
Jun 26 11:37:00 kali ModemManager[1026]: <msg> [base-manager] couldn't check support for device '/sys/devices/pci0000:00/0000:00:03.0': not supported by any plugin
Jun 26 11:37:00 kali ModemManager[1026]: <msg> [base-manager] couldn't check support for device '/sys/devices/pci0000:00/0000:00:08.0': not supported by any plugin

(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$
```

10 – Disable ModemManager:

Stopped and disabled the ModemManager service.



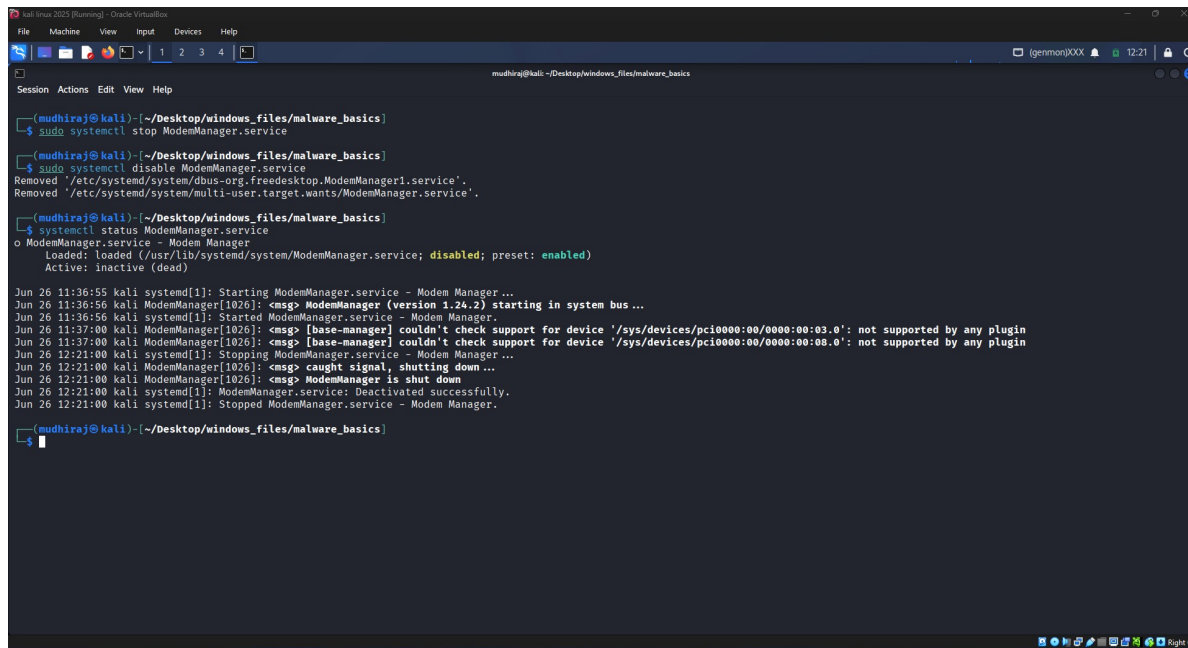
```
(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$ sudo systemctl stop ModemManager.service

(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$ sudo systemctl disable ModemManager.service
Removed /etc/systemd/system/dbus-org.freedesktop.ModemManager1.service'.
Removed /etc/systemd/system/multi-user.target.wants/ModemManager.service'.

(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$
```

11 – Verification:

Verified ModemManager is disabled and inactive.



```
(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$ sudo systemctl stop ModemManager.service

(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$ sudo systemctl disable ModemManager.service
Removed /etc/systemd/system/dbus-org.freedesktop.ModemManager1.service'.
Removed /etc/systemd/system/multi-user.target.wants/ModemManager.service'.

(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$ systemctl status ModemManager.service
o ModemManager.service - Modem Manager
   Loaded: loaded (/usr/lib/systemd/system/ModemManager.service; disabled; preset: enabled)
   Active: inactive (dead)

Jun 26 11:36:55 kali systemd[1]: Starting ModemManager.service - Modem Manager...
Jun 26 11:36:56 kali ModemManager[1026]: <msg> ModemManager (version 1.24.2) starting in system bus ...
Jun 26 11:36:56 kali systemd[1]: Started ModemManager.service - Modem Manager.
Jun 26 11:37:00 kali ModemManager[1026]: <msg> [base-manager] couldn't check support for device '/sys/devices/pci0000:00/0000:00:03.0': not supported by any plugin
Jun 26 11:37:00 kali ModemManager[1026]: <msg> [base-manager] couldn't check support for device '/sys/devices/pci0000:00/0000:00:08.0': not supported by any plugin
Jun 26 12:21:00 kali systemd[1]: Stopping ModemManager.service - Modem Manager...
Jun 26 12:21:00 kali ModemManager[1026]: <msg> caught signal, shutting down ...
Jun 26 12:21:00 kali ModemManager[1026]: <msg> ModemManager is shut down
Jun 26 12:21:00 kali systemd[1]: ModemManager.service: Deactivated successfully.
Jun 26 12:21:00 kali systemd[1]: Stopped ModemManager.service - Modem Manager.

(mudhiraj@kali)-[~/Desktop/windows_files/malware_basics]
$
```

Conclusion:

The Kali Linux virtual machine was successfully hardened by applying package updates, cleaning unused packages, configuring basic iptables firewall rules to allow SSH and block Telnet, and disabling the unused ModemManager service. These actions reduce the attack surface and improve the system's security posture.